

SUPPLEMENTARY UNITS	SYMBOL	DEFINITION
Weber	Wb	A unit of magnetic flux that produces an electromagnetic force of 1 volt when the flux is reduced to zero at a uniform rate of 1 second.
Tesla	T	A unit of magnetic flux density, the equivalent of 1 weber of magnetic flux per meter squared.
Henry	H	A unit of inductance in which an induced electromotive force of 1 volt is produced when the current is varied at the rate of 1 ampere per second.
Degree Celsius	°C	A unit of temperature on the centigrade scale, with the ice point at 0°C and boiling point at 100°C.
Lumen	lm	A unit of luminous flux equal to the amount of light given out through a solid angle of 1 steradian by a point source of 1 candela intensity radiating uniformly in all directions.
Lux	lx	A unit of illumination that describes the illumination given by 1 lumen over an area of 1 square meter.
Becquerel	Bq	A unit of radioactivity. One becquerel describes the activity of radioactive material in which one nucleus decays per second.
Gray	Gy	A unit of an absorbed dose of ionizing radiation and the energy it gives off. It is equal to the absorption of 1 joule per kilogram of irradiated material.
Sievert	Sv	A unit of the dose equivalent needed for protection against ionizing radiation.
Katal	kat	A unit that measures the activity, or property, of catalysts such as enzymes. For example, 1 katal of trypsin is the amount needed to break a mole of peptide bonds in 1 second.

COMMON PHYSICAL PROPERTIES

Scientists use a number of symbols when defining processes using mathematical formulae. The following table shows some of the most common physical properties and the symbols that are used to represent them. The units by which the different properties are measured are indicated by their SI units together with the relevant symbols.

PHYSICAL PROPERTY	SYMBOL	SI UNIT	SI UNIT SYMBOL
Acceleration, deceleration	a	meter/second ² kilometer/hour/second	m s ⁻² km h ⁻¹ s ⁻¹
Angular velocity	ω	radian/second	rad s ⁻¹
Density	ρ	kilogram/meter ³ kilogram/milliliter	kg m ⁻³ kg mL ⁻¹
Electric charge	Q, q	coulomb	C
Electric current	I, i	ampere (coulomb/second)	A (C s ⁻¹)

PHYSICAL PROPERTY	SYMBOL	SI UNIT	SI UNIT SYMBOL
Electrical energy	–	megajoule kilowatt-hour	MJ kWh
Electrical power	P	watt (joule/second)	W (J s ⁻¹)
Electromotive force (EMF)	E	volt (watt/ampere)	V (W A ⁻¹)
Electrical conductance	S	siemen (ohm ⁻¹)	A V ⁻¹
Electrical resistance	R	ohm (volt/ampere)	Ω (V A ⁻¹)
Frequency	f	hertz (cycles/second)	Hz (s ⁻¹)
Force	F	newton (kilogram meter/second ²)	N (kg m s ⁻²)
Gravitational intensity, field strength	–	newton/kilogram	N kg ⁻¹
Magnetic field strength	H	ampere/meter	A m ⁻¹
Magnetic flux	Φ	weber	Wb
Magnetic flux density	B	tesla (weber/meter ²)	T (Wb m ⁻²)
Mass	m	kilogram	kg
Mechanical power	P	watt (joule/second)	W (J s ⁻¹)
Moment of inertia	I	kilogram meter ²	kg m ²
Momentum	p	kilogram meter/second	kg m s ⁻¹
Pressure	P	pascal (newton/meter ²)	Pa (N m ⁻²)
Quantity of substance	n	mole	mol
Specific heat capacity	C or c	joule/kilogram/kelvin	J kg ⁻¹ K ⁻¹
Specific latent heats of fusion, vaporization	L	joule/kilogram	J kg ⁻¹
Torque	τ	newton meter	N m
Velocity, speed	u, v	meter/second kilometer/hour	m s ⁻¹ km h ⁻¹
Volume	V	meter ³ milliliter	m ³ mL
Wavelength	λ	meter	m
Weight	W	newton	N
Work, energy	W	joule (newton meter)	J (N m)